1 Introduction

The multiple linear regression model

For $i = 1, \dots, n$ observations and k regressors,

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \varepsilon_i$$

In matrix form, with $p = k + 1$ parameters,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$$

- $\mathbf{y}$ is $n \times 1$, $\mathbf{X}$ is $n \times p$, $\boldsymbol{\beta}$ is $p \times 1$
- The normality assumption is what makes exact t and F inference possible

OLS estimator and its sampling distribution

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$
$$\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta},\ \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$$

- Write C_{jj} for the jth diagonal element of $(\mathbf{X}'\mathbf{X})^{-1}$, so $\mathrm{Var}(\hat{\beta}_j) = \sigma^2 C_{jj}$
- σ^2 is unknown, so we must estimate it

Estimating σ^2

$$\hat{\sigma}^2 = \mathrm{MS}_{\mathrm{Res}} = \frac{\mathrm{SS}_{\mathrm{Res}}}{n-p}, \qquad \frac{(n-p)\,\mathrm{MS}_{\mathrm{Res}}}{\sigma^2} = \frac{\mathrm{SS}_{\mathrm{Res}}}{\sigma^2} \sim \chi^2_{n-p}$$

- $\mathrm{MS}_{\mathrm{Res}}$ is independent of $\hat{\boldsymbol{\beta}}$
- Normal divided by an independent $\sqrt{\chi^2/\mathrm{df}}$ gives a t distribution

Running example: delivery time data (MPV Example 3.1)

A route driver restocks vending machines. Model the delivery time y (minutes) using x_1 = number of cases stocked and x_2 = distance walked (feet), $n = 25$.

Obs	y (min)	x_1 (cases)	x_2 (ft)
1	16.68	7	560
2	11.50	3	220
3	12.03	3	340
4	14.88	4	80
5	13.75	6	150
⋮	⋮	⋮	⋮
25	10.75	4	150

Explain this computer output

$$\hat{y} = 2.3412 + 1.6159\,x_1 + 0.01438\,x_2$$

```
Coefficients:
               Estimate   Std. Error   t value Pr(>|t|)
(Intercept)    2.341231     1.096730     2.135 0.044170
cases          1.615907     0.170735     9.464 3.25e-09
distance       0.014385     0.003613     3.981 0.000631

Residual standard error:   3.259 on 22 degrees of freedom
Multiple R-squared:  0.9596,  Adjusted R-squared:  0.9559
F-statistic: 261.2 on 2 and 22 DF,     p-value: 4.687e-16
```

2 Test for Significance of Regression

Question

We have fitted a model with regressors $x_1, \ldots, x_k$. How do we decide whether *any* of them is related to y *in the population*?

Hypotheses for the overall test

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$$
$$H_1 : \beta_j \neq 0 \text{ for at least one } j$$

- Under H_0 the model reduces to $y = \beta_0 + \varepsilon$
- The intercept β_0 is *not* part of the hypothesis

Decomposing the total variability

$$\mathrm{SS_T} = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \mathbf{y}'\mathbf{y} - n\bar{y}^2$$
$$\mathrm{SS_R} = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} - n\bar{y}^2$$
$$\mathrm{SS_{Res}} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y}$$
$$\mathrm{SS_T} = \mathrm{SS_R} + \mathrm{SS_{Res}}$$

Degrees of freedom and mean squares

- $\mathrm{SS_T}$ has $n - 1$ degrees of freedom
- $\mathrm{SS_R}$ has k degrees of freedom (one per slope)

- $\mathrm{SS}_{\mathrm{Res}}$ has $n - p = n - k - 1$ degrees of freedom

$$\mathrm{MS}_{\mathrm{R}} = \frac{\mathrm{SS}_{\mathrm{R}}}{k}, \qquad \mathrm{MS}_{\mathrm{Res}} = \frac{\mathrm{SS}_{\mathrm{Res}}}{n-p}$$

Distributional results

Under H_0 and normal errors,

$$\frac{\mathrm{SS}_{\mathrm{R}}}{\sigma^2} \sim \chi^2_k, \qquad \frac{\mathrm{SS}_{\mathrm{Res}}}{\sigma^2} \sim \chi^2_{n-p}, \qquad \mathrm{SS}_{\mathrm{R}} \perp \mathrm{SS}_{\mathrm{Res}}$$

- Under H_1, $\mathrm{SS}_{\mathrm{R}}/\sigma^2$ has a noncentral χ^2_k distribution
- We state these results without proof

The F statistic

$$F_0 = \frac{\mathrm{SS}_{\mathrm{R}}/k}{\mathrm{SS}_{\mathrm{Res}}/(n-p)} = \frac{\mathrm{MS}_{\mathrm{R}}}{\mathrm{MS}_{\mathrm{Res}}} \sim F_{k,\,n-p} \quad \text{under } H_0$$

Reject H_0 at level α if $F_0 > F_{\alpha,\,k,\,n-p}$, or equivalently if the p-value $P(F_{k,n-p} > F_0)$ is below α.

Why a large F_0 is evidence against H_0

Let $\boldsymbol{\beta}^* = (\beta_1, \ldots, \beta_k)'$ and let $\mathbf{X}_c$ be the regressor matrix with columns centered at their means.

$$\mathrm{E}[\mathrm{MS}_{\mathrm{Res}}] = \sigma^2$$
$$\mathrm{E}[\mathrm{MS}_{\mathrm{R}}] = \sigma^2 + \frac{\boldsymbol{\beta}^{*\prime}\mathbf{X}_c'\mathbf{X}_c\boldsymbol{\beta}^*}{k}$$

- Under H_0 both mean squares estimate σ^2, so F_0 is near 1
- Under H_1 the extra term is positive, so F_0 tends to be large
- The test is one-sided: only large values of F_0 reject

The ANOVA table

Source	Sum of squares	df	Mean square	F_0
Regression	SS_{R}	k	MS_{R}	$\mathrm{MS}_{\mathrm{R}}/\mathrm{MS}_{\mathrm{Res}}$
Residual	$\mathrm{SS}_{\mathrm{Res}}$	$n-p$	$\mathrm{MS}_{\mathrm{Res}}$	
Total	SS_{T}	$n-1$		

ANOVA table: delivery time data

Source	Sum of squares	df	Mean square	F_0	p-value
Regression	5550.81	2	2775.41	261.24	4.7×10^{-16}
Residual	233.73	22	10.62		
Total	5784.54	24			

Decision for the delivery time data

- Critical value: $F_{0.05,\,2,\,22} = 3.44$
- Observed: $F_0 = 261.24 \gg 3.44$, with $p \approx 4.7 \times 10^{-16}$
- Reject H_0: at least one of cases and distance is linearly related to delivery time
- The test does not say *which* regressor(s) matter

3 Coefficient of Determination

R^2 and adjusted R^2

$$R^2 = \frac{\mathrm{SS_R}}{\mathrm{SS_T}} = 1 - \frac{\mathrm{SS_{Res}}}{\mathrm{SS_T}}$$
$$R^2_{\mathrm{adj}} = 1 - \frac{\mathrm{SS_{Res}}/(n-p)}{\mathrm{SS_T}/(n-1)}$$

- Delivery time data: $R^2 = 5550.81/5784.54 = 0.9596$ and $R^2_{\mathrm{adj}} = 0.9559$
- R^2 never decreases when a regressor is added; R^2_{adj} can

F_0 as a function of R^2

$$F_0 = \frac{R^2/k}{(1-R^2)/(n-p)}$$

- Delivery time data: $F_0 = \dfrac{0.9596/2}{0.0404/22} = 261.2$
- For fixed k and n, F_0 is increasing in R^2: the overall test asks whether R^2 is larger than chance alone would produce

4 Tests on Individual Coefficients

Question

In the simple regression of y on x_1 alone, the $\widehat{beta}_1$ is 2.176. In the multiple regression on x_1 and x_2, it is 1.616 ($\hat{\beta}_2$ is 0.014). Why do they differ?

Hypotheses for an individual coefficient

$$H_0 : \beta_j = 0$$
$$H_1 : \beta_j \neq 0$$

- This is a *marginal* (partial) test: it assesses the contribution of x_j given that all the other regressors are already in the model
- Changing the other regressors in the model changes the test

Standard error of $\hat{\beta}_j$

$$\begin{aligned}\mathrm{Var}(\hat{\beta}_j) &= \sigma^2 C_{jj} \\ \mathrm{se}(\hat{\beta}_j) &= \sqrt{\hat{\sigma}^2 C_{jj}} = \sqrt{\mathrm{MS}_{\mathrm{Res}}\, C_{jj}}\end{aligned}$$

where C_{jj} is the jth diagonal element of $(\mathbf{X}'\mathbf{X})^{-1}$.

The t statistic

$$t_0 = \frac{\hat{\beta}_j}{\mathrm{se}(\hat{\beta}_j)} \sim t_{n-p} \quad \text{under } H_0$$

Reject H_0 at level α if $|t_0| > t_{\alpha/2,\, n-p}$, or if the two-sided p-value $2\,P(t_{n-p} > |t_0|)$ is below α.

Testing a general null value

For $H_0 : \beta_j = \beta_{j0}$,

$$t_0 = \frac{\hat{\beta}_j - \beta_{j0}}{\mathrm{se}(\hat{\beta}_j)} \sim t_{n-p}$$

- One-sided alternatives $H_1 : \beta_j > \beta_{j0}$ or $\beta_j < \beta_{j0}$ use $t_{\alpha,\, n-p}$ and a one-sided p-value

Delivery time data: test for cases (β_1)

From $(\mathbf{X}'\mathbf{X})^{-1}$ we get $C_{11} = 0.002744$, and $\mathrm{MS}_{\mathrm{Res}} = 10.6242$.

$$\begin{aligned}\mathrm{se}(\hat{\beta}_1) &= \sqrt{10.6242 \times 0.002744} = 0.1707 \\ t_0 &= \frac{1.6159}{0.1707} = 9.46\end{aligned}$$

$t_{0.025,\,22} = 2.074$, so $|t_0| > 2.074$ and we reject $H_0 : \beta_1 = 0$.

Delivery time data: all coefficients

Term	Estimate	Std. error	t_0	p-value
Intercept	2.3412	1.0967	2.135	0.0442
Cases (x_1)	1.6159	0.1707	9.464	3.3×10^{-9}
Distance (x_2)	0.014385	0.003613	3.981	0.00063

At $\alpha = 0.05$ each term is significant given the other regressors.

Interpreting the slopes

- Each additional case increases mean delivery time by 1.62 minutes, holding distance fixed
- Each additional 100 ft of distance increases mean delivery time by 1.44 minutes, holding cases fixed
- "Holding the other regressor fixed" is the whole content of a partial coefficient

Interpreting the intercept

- $\hat{\beta}_0 = 2.34$ is the fitted mean time for a delivery with 0 cases and 0 ft of distance
- The data have $x_1 \in [2, 30]$ and $x_2 \in [36, 1460]$, so the origin is outside the region of the data
- Treat $\hat{\beta}_0$ as a term that positions the fitted plane, not as a physical quantity

5 Confidence Intervals on Coefficients

The pivotal quantity

$$\frac{\hat{\beta}_j - \beta_j}{\text{se}(\hat{\beta}_j)} \sim t_{n-p}$$

$$P\left(-t_{\alpha/2,\, n-p} \le \frac{\hat{\beta}_j - \beta_j}{\text{se}(\hat{\beta}_j)} \le t_{\alpha/2,\, n-p}\right) = 1 - \alpha$$

Confidence interval for β_j

Solving the inequality for β_j:

$$\hat{\beta}_j \pm t_{\alpha/2,\, n-p}\ \text{se}(\hat{\beta}_j)$$

- Interpretation: in repeated sampling, $100(1-\alpha)\%$ of intervals built this way cover the true β_j
- It is a statement about the procedure, not a probability statement about one fixed interval

Delivery time data: 95% intervals

Example for cases: $1.6159 \pm 2.074 \times 0.1707 = (1.262,\ 1.970)$

Term	Lower	Upper
Intercept	0.0668	4.6157
Cases (x_1)	1.2618	1.9700
Distance (x_2)	0.00689	0.02188

6 Confidence Interval on the Mean Response

Question

A new route has $x_1 = 8$ cases and $x_2 = 275$ ft. What is the difference between (a) estimating the *average* delivery time over all such routes and (b) predicting the time of *one particular* such delivery?

Point estimate at $\mathbf{x}_0$

Let $\mathbf{x}_0 = (1, x_{01}, \ldots, x_{0k})'$. Then

$$\hat{y}_0 = \mathbf{x}_0' \hat{\boldsymbol{\beta}}$$

estimates both $\text{E}[y \mid \mathbf{x}_0]$ and the new observation y_0.

$$\mathbf{x}_0 = (1, 8, 275)' \ \Rightarrow\ \hat{y}_0 = 2.3412 + 1.6159(8) + 0.014385(275) = 19.22 \text{ min}$$

Variance of $\hat{y}_0$ and its distribution

$$\mathrm{Var}(\hat{y}_0) = \sigma^2\, \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$$

$$\frac{\hat{y}_0 - \mathrm{E}[y \mid \mathbf{x}_0]}{\sqrt{\mathrm{MS}_{\mathrm{Res}}\, \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}} \sim t_{n-p}$$

Confidence interval on the mean response

$$\hat{y}_0 \pm t_{\alpha/2,\, n-p}\sqrt{\mathrm{MS}_{\mathrm{Res}}\, \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}$$

Delivery time data at $\mathbf{x}_0 = (1, 8, 275)'$
Here $\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = 0.05397$.

$$\text{half-width} = 2.074\sqrt{10.6242 \times 0.05397} = 1.570$$
$$95\%\ \text{CI} = 19.22 \pm 1.57 = (17.65,\ 20.79)$$

7 Prediction of New Observations

The forecast error for a new observation

$$y_0 = \mathbf{x}_0'\boldsymbol{\beta} + \varepsilon_0, \qquad \varepsilon_0 \sim N(0, \sigma^2), \quad \varepsilon_0 \perp \hat{\boldsymbol{\beta}}$$

$$y_0 - \hat{y}_0 \sim N(0,\ \sigma^2\,[1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0])$$

The extra σ^2 is the variance of the new observation's own noise ε_0.

Prediction interval

$$\hat{y}_0 \pm t_{\alpha/2,\, n-p}\sqrt{\mathrm{MS}_{\mathrm{Res}}\,[1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0]}$$

$$\text{Delivery time data: half-width} = 2.074\sqrt{10.6242 \times 1.05397} = 6.94$$
$$95\%\ \text{PI} = 19.22 \pm 6.94 = (12.28,\ 26.16)$$

Confidence interval versus prediction interval

	Target	Variance term inside $\sqrt{\cdot}$	Delivery data at $\mathbf{x}_0$
CI	$\mathrm{E}[y \mid \mathbf{x}_0]$	$\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$	(17.65, 20.79)
PI	y_0	$1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$	(12.28, 26.16)

Question

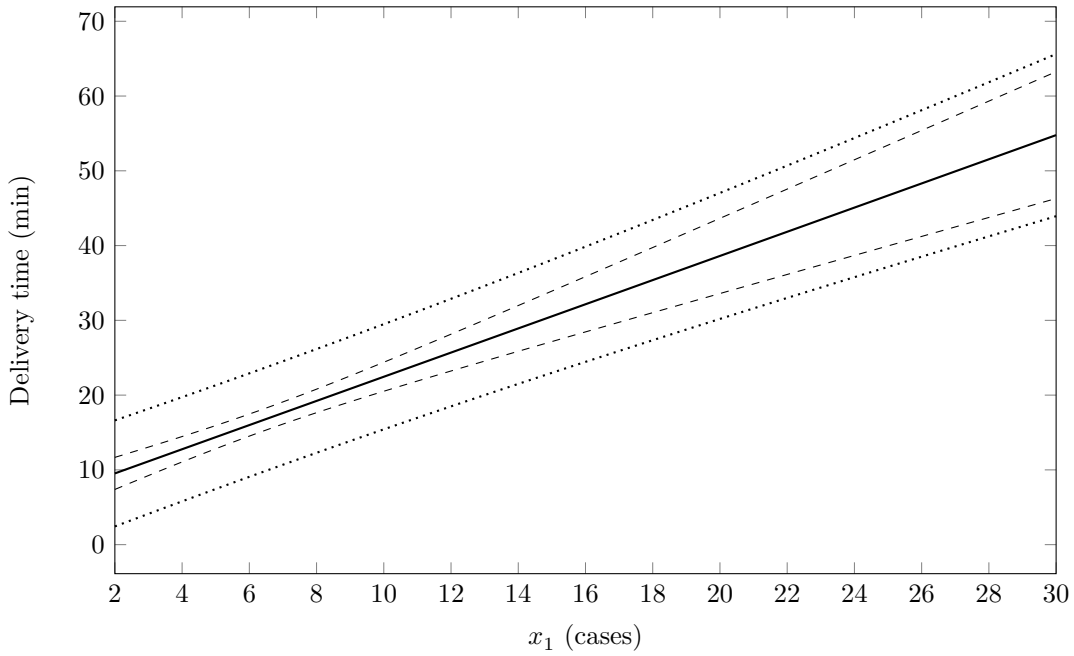
As $n \to \infty$, what happens to the width of the CI on the mean response, and to the width of the PI?

- $\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 \to 0$, so the CI shrinks to a point: we learn the mean exactly
- The PI does not shrink to a point: it tends to $\hat{y}_0 \pm z_{\alpha/2}\sigma$, because a new observation still carries its own noise ε_0

CI and PI bands

Slice of the fitted plane at $x_2 = 275$ ft, as a function of x_1.

- Inner band: CI on the mean
- Outer band: PI for a new observation
- Both bands are narrowest near the center of the data and widen away from it



Prediction in R

```
new <- data.frame(cases = 8, distance = 275)
predict(fit, new, interval = "confidence")
predict(fit, new, interval = "prediction")
```

```
       fit      lwr      upr
  19.22432 17.65390 20.79474
  19.22432 12.28456 26.16407
```

The first output is the CI on the mean response, the second is the PI.

8 Hidden Extrapolation

Question

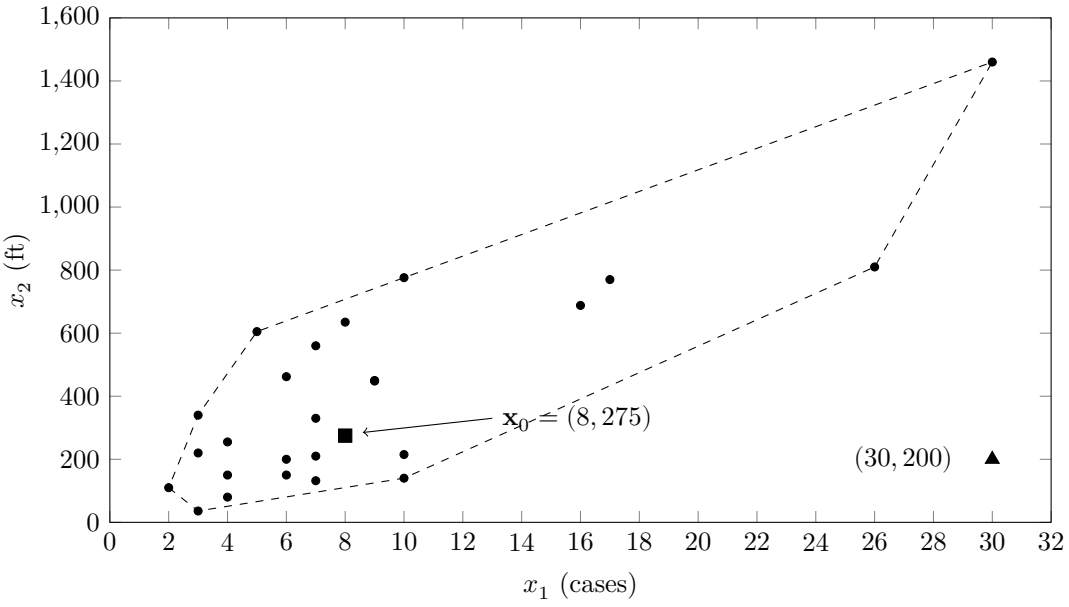
Delivery time data: cases range over $[2, 30]$ and distance over $[36, 1460]$. Is a route with $x_1 = 30$ cases and $x_2 = 200$ ft safe to predict for?

- Each coordinate is inside its own observed range, but the *combination* is not near any observed delivery
- The large-case deliveries all have long distances; the model has no information about a 30-case, short-distance route

Regressor variable hull

Define $h_{00} = \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$ and let $h_{\max}$ be the largest diagonal element of the hat matrix $\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

- $h_{00} \le h_{\max}$: $\mathbf{x}_0$ lies inside the region covered by the data
- $h_{00} > h_{\max}$: hidden extrapolation
- Dashed polygon: convex hull of the observed (x_1, x_2) pairs



Checking the two candidate routes

Delivery time data: $h_{\max} = 0.498$ (observation 9).

(x_{01}, x_{02})	h_{00}	$h_{\max}$	Conclusion
$(8, 275)$	0.054	0.498	inside the data region
$(30, 200)$	1.757	0.498	hidden extrapolation

For the second route, the CI and PI formulas still compute a number, but the model gives no evidence the fit holds there.

9 Summary

Summary of procedures

Procedure	Target	Statistic or interval	Reference
Overall test	$H_0 : \beta_1 = \cdots = \beta_k = 0$	$F_0 = \mathrm{MS_R}/\mathrm{MS_{Res}}$	$F_{k,\,n-p}$
t test	$H_0 : \beta_j = \beta_{j0}$	$t_0 = \dfrac{\hat{\beta}_j - \beta_{j0}}{\mathrm{se}(\hat{\beta}_j)}$	t_{n-p}
CI for β_j	β_j	$\hat{\beta}_j \pm t_{\alpha/2,n-p}\,\mathrm{se}(\hat{\beta}_j)$	t_{n-p}
CI for mean	$\mathrm{E}[y \mid \mathbf{x}_0]$	$\hat{y}_0 \pm t_{\alpha/2,n-p}\sqrt{\mathrm{MS_{Res}}\,h_{00}}$	t_{n-p}
PI	y_0	$\hat{y}_0 \pm t_{\alpha/2,n-p}\sqrt{\mathrm{MS_{Res}}(1+h_{00})}$	t_{n-p}

with $h_{00} = \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$.